

Deepak Mallampati

FORWARD DEPLOYED / INFRASTRUCTURE ENGINEER

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US-based (F-1 OPT); open to UK relocation

PROFESSIONAL SUMMARY

Infrastructure / forward-deployed engineer with 4+ years delivering production AI systems into **regulated enterprise environments** (banking, healthcare). Strong in cloud infrastructure and IaC - AWS, Terraform, Docker/Kubernetes, CI/CD - with deployments hardened for **HIPAA, SOC 2, and PCI DSS** compliance, audit logging, RBAC, and zero-trust controls. Built fault-tolerant orchestration that cut failed production runs by **60%** and observability frameworks that catch regressions before release. Comfortable navigating security reviews and change-management gates, and translating enterprise constraints into reliable, repeatable deployments.

CORE COMPETENCIES

AWS • Terraform / IaC • Docker & Kubernetes • Cloud Networking • CI/CD • Security & Compliance (HIPAA/SOC2/PCI) • Observability & Monitoring • LLM Infrastructure • Deployment Automation • Reliability Engineering

Infrastructure & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps) | Model Monitoring & Observability

Security & Compliance: HIPAA, SOC 2, PCI DSS | OAuth 2.0, JWT, RBAC | PII Masking/Redaction | Audit Logging | Zero-Trust Architecture

Languages & Backend: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | Git

AI/ML: LLM/RAG production systems | Agentic orchestration | LLM eval (RAGAS, BERTScore) | Vector search (FAISS, Pinecone, Weaviate)

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Built fault-tolerant **automation infrastructure** with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.
- Designed and operationalized **privacy-preserving, compliant data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, differential privacy) with zero PII exposure.
- Established an LLM evaluation/monitoring framework (RAGAS, BERTScore) with latency and accuracy dashboards to surface regressions before release.
- Architected production-grade agentic orchestration on a conductor-subagent framework, cutting token consumption by 42% while sustaining accuracy.
- Optimized inference cost/latency via prompt compression, semantic caching, and model routing, sustaining SLA targets.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms**; partnered across product, platform, and data engineering teams.
- Delivered scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams, integrating with production data pipelines and observability tooling.
- Engineered safeguards and content controls for secure, policy-compliant AI responses, reducing unsafe outputs by 42%.
- Worked within a regulated financial-services environment with strict security and governance requirements.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Reduced record-linkage re-identification risk from 3.38% to 0.32% via k-anonymity + l-diversity while retaining 99% of baseline predictive performance.
- Fine-tuned Qwen3-27B via 4-bit QLoRA on an NVIDIA H100 with curated instruction data.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Built regression/classification models on operational pipeline-storage data; engineered end-to-end preprocessing pipelines for equipment-failure prediction and anomaly detection.

PROJECTS

career-ops

Autonomous AI job-search pipeline with scheduling, orchestration, and reliability built in - scans listings, scores fit, and generates tailored applications overnight.

MangaLens

Shipped Chrome extension (live on Chrome Web Store) delivering real-time AI translation across 29+ sites.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0